MATH 110 — Week 7

Notes Chapters 12–13; Homework 7

§1 Upper-triangular matrices

The question. How does a matrix record a nested family of invariant subspaces? For $A = \left(\begin{smallmatrix} 1 & 2 \\ 0 & 3 \end{smallmatrix}\right)$, the first coordinate line is invariant because $Ae_1 = e_1$.

Tools from earlier weeks.

- Week 2: extend an independent list to a basis by adding vectors outside its span; basis length defines dimension and an independent list of full length is a basis.
- Weeks 3–4: rank-nullity follows by extending a kernel basis and using the remaining images as a range basis; thus an operator on finite dimension is injective, surjective and invertible under the same conditions.
- Weeks 4–5: the root bound follows by successively removing root factors; $\dim(V/U) = \dim V - \dim U$ follows from an adapted basis; evaluating a polynomial on T/U sends $v + U$ to $p(T)v + U$, by induction on powers.
- Week 6: for $p(x) = \sum_j a_j x^j$, define $p(T) = \sum_j a_j T^j$ with $T^0 = I$. Eigenvectors for distinct eigenvalues are independent by applying a shifted operator to a shortest dependence; polynomial evaluation preserves products by distributing finite sums of powers, so polynomials in T commute.
- Week 6: $\|x\|_1 = \sum_i |x_i|$ and a column-stochastic P satisfy $\|Px\|_1 \le \sum_{i,j} P_{ij}|x_j| = \|x\|_1$, by the triangle inequality and column sums 1.
- Week 6: every nonzero finite-dimensional complex operator has an eigenvalue by factoring a relation among vector iterates and using the fundamental theorem of algebra. That theorem was proved there using the explicitly cited extreme-value theorem; its repeated root extraction gives complex factorization.

Definition (Notes Definition 33; LADR 5.37–5.38, 5.48)**.** A square matrix $A = (a_{ij})$ is upper triangular when $a_{ij} = 0$ for $i > j$, and diagonal when $a_{ij} = 0$ for $i \neq j$. The main diagonal consists of the entries a_{jj}.

Example (Notes Example 53)

The matrix $\left(\begin{smallmatrix} 1 & 2 & 3 & 4 \\ 0 & 5 & 6 & 7 \\ 0 & 0 & 8 & 9 \\ 0 & 0 & 0 & 10 \end{smallmatrix}\right)$ is upper triangular, with main diagonal $1, 5, 8, 10$. It is not diagonal because, for example, $a_{12} = 2 \neq 0$.

Proposition (Notes Proposition 59; LADR 5.39)

For a basis $v_1, \dots, v_n$, the following are equivalent: the matrix of T is upper triangular; $Tv_j \in \operatorname{span}(v_1, \dots, v_j)$ for each j; every initial span $U_j = \operatorname{span}(v_1, \dots, v_j)$ is invariant.

Where the proof comes from

Want. Recognize triangular form without computing every entry.
Know. Matrix column j gives the coordinates of Tv_j (Week 3).
Key step. Coordinates with indices greater than j vanish exactly when $Tv_j \in U_j$.

Proof. 1. The coordinate observation proves equivalence of the first two conditions.
2. If $Tv_j \in U_j$, then for $k \le j$, $Tv_k \in U_k \subseteq U_j$; linearity proves $T(U_j) \subseteq U_j$.
3. If U_j is invariant, $v_j \in U_j$ gives $Tv_j \in U_j$.

□

Proposition (Notes Proposition 60; LADR 5.41)

The eigenvalues of an upper-triangular matrix are exactly its diagonal entries, viewed as elements of the underlying field.

Where the proof comes from

Want. Solve the eigenvalue test using only diagonal entries.
Know. Week 6 equates eigenvalues with noninvertibility of $T - \lambda I$.
Key step. An upper-triangular matrix is invertible exactly when every diagonal entry is nonzero.

Proof. 1. If all diagonal entries of B are nonzero, solve $Bx = y$ from the last row upward: each row determines its new coordinate uniquely by division by b_{jj}. Thus B is invertible.
2. If some diagonal entry is zero, let j be the first such index. Put $x_j = 1$ and $x_k = 0$ for $k > j$; solve the first $j - 1$ equations upward in reverse order using their nonzero diagonal entries. Then $Bx = 0$ with $x \neq 0$.
3. Apply these facts to $B = A - \lambda I$, whose diagonal entries are $a_{jj} - \lambda$.

□

Example (Notes Example 54) 1. On $\mathcal{P}_3(F)$ for $F = \mathbb{R}$ or $\mathbb{C}$, the power rule gives differentiation matrix $\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ in $1, x, x^2, x^3$.
2. Its only eigenvalue is 0 by Proposition 60. Its eigenspace is the constants because $Dp = 0$ exactly when every positive-degree coefficient is zero.

$$x^3 \xrightarrow{D} 3x^2 \xrightarrow{D} 6x \xrightarrow{D} 6 \xrightarrow{D} 0$$

Differentiation sends each polynomial to a lower-degree initial span.

Proposition (Notes Proposition 61; LADR 5.47)

Every operator on a finite-dimensional complex vector space has an upper-triangular matrix in some basis.

Where the proof comes from

Want. Construct a basis with invariant initial spans.
Know. Every nonzero finite-dimensional complex space has an eigenvector (Week 6).
Key step. Use the proper invariant range of $T - \lambda I$, then extend its triangular basis.

Proof. 1. Induct on $n = \dim V$; the empty basis handles $n = 0$. For $n > 0$, choose an eigenvalue λ by Week 6.
2. Put $U = \operatorname{range}(T - \lambda I)$. It is proper because $T - \lambda I$ is not surjective; it is invariant because it commutes with T.
3. By induction, take a triangular basis $v_1, \dots, v_m$ for $T|_U$, then extend it to a basis of V (Week 2).
4. For $j > m$, $Tv_j = (T - \lambda I)v_j + \lambda v_j \in U + Fv_j \subseteq U_j$. For $j \le m$ the induction hypothesis gives the same containment. Apply Proposition 59.

□

Shortcuts

- Triangular eigenvalues are the diagonal entries, because subtracting λI changes only that diagonal.
- To prove triangular form, check initial-span invariance on basis vectors.

Traps

- A triangular matrix need not have a basis of eigenvectors: $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ has only the eigenvector line Fe_1.
- Real triangularization can fail: the real quarter-turn has no real eigenvalue, whereas every nonempty real triangular matrix has a diagonal eigenvalue.

§2 Diagonalization and repeated application

The question. When can repeated application be reduced to scalar powers? For $\operatorname{diag}(2,3)$, its kth power sends (a,b) to $(2^k a, 3^k b)$.

Definition (Notes Definition 34; LADR 5.50, 5.52)**.** An operator is diagonalizable over F if its matrix in some basis is diagonal. An eigenbasis is a basis of eigenvectors. The geometric multiplicity of an eigenvalue λ is $\dim E(\lambda, T)$.

Proposition (Notes Proposition 62; LADR 5.54–5.55)

For any distinct eigenvalues, their eigenspace sum is direct and $\sum_j \dim E(\lambda_j, T) \le \dim V$. For the complete list of distinct eigenvalues $\lambda_1, \dots, \lambda_r$ of an operator on finite-dimensional V, the following are equivalent: T is diagonalizable; V has an eigenbasis; $V = \bigoplus_{j=1}^{r} E(\lambda_j, T)$; $\dim V = \sum_j \dim E(\lambda_j, T)$.

Where the proof comes from

Want. Decide whether the available eigenspaces provide a full basis.
Know. Eigenvectors with distinct eigenvalues are independent (Week 6).
Key step. First prove the eigenspace sum is direct, then compare dimensions.

Proof. 1. If $w_j \in E(\lambda_j, T)$ and $\sum_j w_j = 0$, the nonzero w_j are eigenvectors with distinct eigenvalues; Week 6 forces every $w_j = 0$. Thus the sum is direct. Its dimension is the sum of the eigenspace dimensions and is at most $\dim V$, by Week 4.

2. A diagonal matrix has $Tv_j = \lambda_j v_j$ for each basis column, and an eigenbasis has a diagonal matrix. This proves the first two conditions equivalent.

3. Concatenating bases of the distinct eigenspaces gives an independent list by step 1. It is a basis precisely when the eigenspaces span V, equivalently when its length equals $\dim V$ (Week 2).

□

Example (LADR 5.49, 5.51, 5.53) 1. A diagonal matrix $\operatorname{diag}(8,5,5)$ in basis v_1, v_2, v_3 has eigenspaces $E(8,T) = Fv_1$ and $E(5,T) = \operatorname{span}(v_2, v_3)$ over $\mathbb{R}$ or $\mathbb{C}$, by solving the coordinate equations.

2. For $A = \left(\begin{smallmatrix} 41 & 7 \\ -20 & 74 \end{smallmatrix}\right)$, direct multiplication gives $A(1,4) = 69(1,4)$ and $A(7,5) = 46(7,5)$.

3. The vectors are independent since $a(1,4) + b(7,5) = 0$ gives $a = -7b$ and $-23b = 0$, hence $a = b = 0$ over $\mathbb{R}$. Thus its matrix in that basis is $\operatorname{diag}(69, 46)$.

Proposition (Notes Proposition 63; LADR 5.58)

An operator on an n-dimensional space with n distinct eigenvalues is diagonalizable.

Where the proof comes from

Want. Avoid solving every eigenspace dimension.
Know. Distinct-eigenvalue eigenvectors are independent.
Key step. Choose one nonzero vector from each of the n eigenspaces.

Proof. 1. Choose $v_j \neq 0$ with $Tv_j = \lambda_j v_j$. Week 6 gives independence.
2. An independent list of length n is a basis (Week 2), hence an eigenbasis. The empty list handles $n = 0$.

□

Example (Notes Example 55; LADR 5.57)
For $n > 1$, let $Tv_1 = 0$ and $Tv_j = v_{j-1}$ for $j > 1$.

1. The matrix is upper triangular with zero diagonal, so its only eigenvalue is 0 by Proposition 60.
2. Solving $T\sum_j a_j v_j = \sum_{j>1} a_j v_{j-1} = 0$ gives $E(0,T) = Fv_1$, of dimension 1.
3. Since $1 < n$, Proposition 62 rules out diagonalizability.

Example (Notes Example 56) 1. For $A = \left(\begin{smallmatrix} 1 & 2 \\ 2 & 3 \end{smallmatrix}\right)$, the Week 6 two-by-two test gives $(t-1)(t-3)-4 = t^2-4t-1$, with roots $2 \pm \sqrt{5}$. Each has eigenvector $(2, 1 \pm \sqrt{5})$, verified by multiplication.
2. Thus A is diagonalizable over $\mathbb{R}$ by Proposition 63. Over $\mathbb{Q}$ it has no eigenvalue, since the roots are irrational, so it has no eigenbasis.
3. For $R = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, the eigenvalues over $\mathbb{C}$ are $i, -i$ with vectors $(1, -i), (1, i)$; over $\mathbb{R}$ there is no eigenvalue.

Proposition (Notes Proposition 64)
If $Tv_j = \lambda_j v_j$ is an eigenbasis and $v = \sum_j a_j v_j$, then $T^k v = \sum_j a_j \lambda_j^k v_j$ for every integer $k \geq 0$. Here $\lambda_j^0 = 1$, including when $\lambda_j = 0$.

Where the proof comes from

Want. Compute high powers without multiplying matrices repeatedly.
Know. T acts as scalar multiplication on each eigenvector line.
Key step. Apply that equation repeatedly and then use linearity.

Proof. 1. Induction on k gives $T^k v_j = \lambda_j^k v_j$, with $k = 0$ the identity equation.
2. Apply T^k to the finite linear combination for v and distribute by linearity.

□

Compute a large power (LADR 5.59) For $A = \left(\begin{smallmatrix} 2 & 1 & 0 \\ 0 & 5 & 3 \\ 0 & 0 & 8 \end{smallmatrix}\right)$:

1. Solve the three eigenvector equations to obtain $v_1 = (1,0,0)$, $v_2 = (1,3,0)$ and $v_3 = (1,6,6)$ for eigenvalues $2, 5, 8$.
2. Solving $e_3 = av_1 + bv_2 + cv_3$ gives $(a,b,c) = (1/6, -1/3, 1/6)$.
3. Proposition 64 gives $A^{100}e_3 = \frac{1}{6}(2^{100} - 2 \cdot 5^{100} + 8^{100},\ 6 \cdot 8^{100} - 6 \cdot 5^{100},\ 6 \cdot 8^{100})$.

Recover the Fibonacci formula (Notes Chapter 12; LADR 5D Exercise 21)

1. Let V be the real sequences satisfying $a_{n+2} = a_{n+1} + a_n$ for $n \geq 0$. Sending a sequence to (a_0, a_1) is an isomorphism: the recurrence determines every later term uniquely.
2. The left shift corresponds to $T(x,y) = (y, x+y)$. Its eigenvalues are $\lambda_\pm = (1 \pm \sqrt{5})/2$, with eigenvectors $(1, \lambda_\pm)$ because $\lambda_\pm^2 = \lambda_\pm + 1$.
3. For the Fibonacci start $(F_0, F_1) = (0,1)$, $(0,1) = ((1,\lambda_+) - (1,\lambda_-))/\sqrt{5}$. The first coordinate after n shifts is therefore $F_n = (\lambda_+^n - \lambda_-^n)/\sqrt{5}$.

Shortcuts

- Solve $Sc = v$ once, where the columns of S are eigenvectors; then replace each c_j by $\lambda_j^k c_j$.
- Distinct eigenvalues certify diagonalizability before the eigenvectors are computed.

Traps

- A repeated eigenvalue may still allow diagonalization: I_2 has only eigenvalue 1 and every basis is an eigenbasis.
- The field matters: the quarter-turn is diagonalizable over $\mathbb{C}$, but has no real eigenbasis.

§3 Stochastic powers and convergence rates

The question. What determines the rate of approach to a stationary distribution? For $P_\epsilon = \left(\begin{smallmatrix} 1-\epsilon & \epsilon \\ \epsilon & 1-\epsilon \end{smallmatrix}\right)$, the difference between the two coordinates is multiplied by $1-2\epsilon$ at each step.

Definition (Notes Chapter 12: eigenmodes and spectral gap)**.** In an eigenbasis expansion, an eigenmode means one component $a_j v_j$. Over $\mathbb{R}$ or $\mathbb{C}$, its coefficient after k iterations has modulus $|a_j|\,|\lambda_j|^k$ by Proposition 64 and multiplicativity of modulus. For nonzero a_j, this tends to zero if $|\lambda_j| < 1$, is constant if $|\lambda_j| = 1$, and tends to infinity if $|\lambda_j| > 1$, by the geometric limits from Week 6 (use reciprocal for growth). For a stochastic operator with a single stationary direction and other eigenvalues of modulus less than 1, its spectral gap is $1 - \max_{\lambda \neq 1} |\lambda|$.

Example (Notes Chapter 12: the three-state chain) 1. Let $P = \left(\begin{smallmatrix} 1/2 & 1/4 & 0 \\ 1/2 & 1/2 & 1/2 \\ 0 & 1/4 & 1/2 \end{smallmatrix}\right)$, $\pi = (1/4, 1/2, 1/4)$, $u = (-1, 0, 1)$ and $w = (1, -2, 1)$. Multiplication gives $P\pi = \pi$, $Pu = u/2$ and $Pw = 0$.

2. These vectors form an eigenbasis because their eigenvalues $1, 1/2, 0$ are distinct. Solving for the initial distribution gives $e_1 = \pi - u/2 + w/4$.
3. For $k \geq 1$, Proposition 64 gives $P^k e_1 = \pi - 2^{-(k+1)} u = (1/4 + 2^{-(k+1)}, 1/2, 1/4 - 2^{-(k+1)})$.
4. The stationary component stays fixed, the 0-component becomes zero after one step, and the remaining error is halved each step. In particular $\|P^k e_1 - \pi\|_1 = 2^{-k}$ for $k \geq 1$.

Example (Notes Chapter 12: a bottleneck) 1. For $0 < \epsilon < 1/2$, $P_\epsilon(1,1) = (1,1)$ and $P_\epsilon(1,-1) = (1-2\epsilon)(1,-1)$.

2. Since $e_1 = \frac{1}{2}(1,1) + \frac{1}{2}(1,-1)$, its kth distribution is $\frac{1}{2}(1,1) + \frac{1}{2}(1-2\epsilon)^k(1,-1)$.
3. Its gap is 2ϵ. Small ϵ makes transitions between the two states rare and the error factor $1-2\epsilon$ close to 1, hence slow convergence.

Proposition (Notes Chapter 12: PageRank eigenvalues)

For the Google matrix $G = \alpha P + (1-\alpha) v \mathbf{1}^\mathsf{T}$ from Week 6, $\mathbb{R}^n = \mathbb{R}r \oplus H$, where $H = \ker \mathbf{1}^\mathsf{T}$, $Gr = r$, and $G|_H = \alpha P|_H$. Every complex eigenvalue of the second restriction has modulus at most α.

Where the proof comes from

Want. Explain why damping controls convergence even when diagonalization fails.

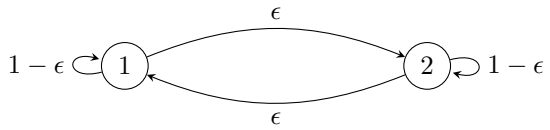
Know. Week 6 proved the invariant decomposition, the restriction identity and the stochastic 1-norm bound.

Key step. Scale the eigenvalue equation on H by α and use the same norm bound.

Proof. 1. The real decomposition and restriction equations are exactly the invariant-decomposition result of Week 6. Over $\mathbb{C}$, use the same equations with complex coordinates and coordinate-sum-zero subspace.

2. If $Gx = \mu x$ for nonzero $x \in H$, then $Px = (\mu/\alpha)x$. Week 6 gives $|\mu/\alpha| \leq 1$, so $|\mu| \leq \alpha$.
3. For probability starts, Week 6 already gives $\|r_k - r\|_1 \leq \alpha^k \|r_0 - r\|_1$ directly; no eigenbasis assumption is required.

□



The two-state matrix retains most mass at its current state when ϵ is small. Its error factor is $1-2\epsilon$.

Shortcuts

- Subtract the stationary vector first: only the other eigenmodes remain, so their moduli determine the error rates when an eigenbasis exists.
- The PageRank 1-norm contraction bound remains valid without diagonalization, by Week 6.

Traps

- Stochasticity alone does not imply convergence: $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ alternates the two point masses.
- A small gap means slow convergence: for $\epsilon = 1/100$, the nonstationary factor is $49/50$, not $1/50$.

§4 The minimal polynomial

The question. Which polynomial relations contain all relations satisfied by an operator? If $T^2 - 4T + 3I = 0$, then every eigenvalue satisfies $\lambda^2 - 4\lambda + 3 = 0$.

Example (LADR 5.20: finite dimension is necessary) 1. On $\mathcal{P}(\mathbb{C})$, define $(Tp)(x) = xp(x)$. This is linear by distributivity of polynomial multiplication.

2. For $p \neq 0$, $\deg(Tp) = \deg p + 1$, whereas a nonzero scalar multiple of p has degree $\deg p$ and $0p = 0 \neq Tp$. Thus T has no eigenvalue. The space is infinite-dimensional because $1, x, x^2, \ldots$ is independent by coefficient comparison.

Definition (Notes Definition 35; LADR 5.14)**.** For $p(x) = \sum_j a_j x^j$, put $p(T) = \sum_j a_j T^j$, with $T^0 = I$. Week 6 proved $(pq)(T) = p(T)q(T)$, linearity of evaluation, and $p(T)v = p(\lambda)v$ when $Tv = \lambda v$.

Definition (Notes Definition 36; LADR 5.21, 5.24)**.** A polynomial is monic if its leading coefficient is 1. For a finite-dimensional operator, its minimal polynomial m_T is the monic polynomial of smallest degree satisfying $m_T(T) = 0$. A nonzero polynomial p satisfying $p(T) = 0$ is called an annihilating polynomial. A polynomial q is divisible by p if $q = ph$ for a polynomial h. On the zero space, $I = 0$ and $m_T = 1$.

Proposition (Existence and uniqueness in Notes Definition 36; LADR 5.22)

The minimal polynomial exists and is unique. If $I, T, \ldots, T^{d-1}$ are independent and $T^d = \sum_{j<d} a_j T^j$, then $m_T(x) = x^d - \sum_{j<d} a_j x^j$.

Where the proof comes from

Want. Obtain a first polynomial relation and make it unique.
Know. $L(V)$ has finite dimension $(\dim V)^2$ (Week 3).
Key step. Use the first dependence of operator powers and normalize its highest coefficient.

Proof. 1. The list $I, T, T^2, \ldots$ is dependent in $L(V)$. A nonzero relation gives an annihilating polynomial; choose its smallest possible degree and divide by its leading coefficient.

2. Two distinct monic polynomials of that degree would have a nonzero difference of smaller degree, also annihilating T, contradicting minimality.

3. Independence of the earlier powers rules out any smaller-degree relation, proving the stated formula. The case $V = \{0\}$ gives the constant polynomial 1.

□

Example (Notes Example 57) 1. On nonzero F^2, $m_I = x - 1$, since $I - I = 0$ and no nonzero constant annihilates I.

2. For $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, $A^2 = \left(\begin{smallmatrix} 5 & 4 \\ 4 & 5 \end{smallmatrix}\right) = 4A - 3I$. Thus $x^2 - 4x + 3$ annihilates A.

3. The matrices I, A are independent because A is not scalar; hence $m_A = x^2 - 4x + 3 = (x-1)(x-3)$.

Proposition (Polynomial division; LADR 4.8)

If $p \neq 0$, every polynomial q has a unique expression $q = ph + r$, where $r = 0$ or $\deg r < \deg p$.

Where the proof comes from

Want. Reduce a polynomial relation to a smaller degree.
Know. Leading coefficients can be divided in a field.
Key step. Repeatedly cancel the leading term; the degree drops each time.

Proof. 1. If $\deg q \geq \deg p$, subtract the unique scalar multiple of $x^{\deg q - \deg p} p$ that cancels its leading term. Record that monomial in h.
2. Repeat until the remainder is zero or has smaller degree than p. The decreasing nonnegative degrees ensure termination.
3. If $ph + r = ph' + r'$, then $p(h - h') = r' - r$. A nonzero left side has degree at least $\deg p$, while the right side has smaller degree; both sides must be zero.

□

Proposition (Notes Proposition 65; LADR 5.29)

For every polynomial q, $q(T) = 0$ if and only if m_T divides q.

Where the proof comes from

Want. Characterize every annihilating polynomial.
Know. Division gives a remainder of degree smaller than m_T.
Key step. Evaluate the division identity and use minimality to force that remainder to vanish.

Proof. 1. If $q = m_T h$, Week 6 polynomial multiplication gives $q(T) = m_T(T)h(T) = 0$.
2. Conversely, write $q = m_T h + r$. If $q(T) = 0$, then $r(T) = 0$; a nonzero r normalized to be monic would contradict minimality. Thus $r = 0$.

□

Proposition (Notes Proposition 66; LADR 5.27(a)–(b))

The eigenvalues of T are exactly the roots of m_T in the underlying field. Over $\mathbb{C}$, $m_T = \prod_j (x - \lambda_j)^{k_j}$ for the distinct eigenvalues λ_j and positive integers k_j (the zero-space product is empty).

Where the proof comes from

Want. Recover eigenvalues from a polynomial relation of minimal degree.
Know. Polynomials act on eigenvectors by scalar evaluation; roots give linear factors (Week 4).
Key step. If a root's shifted operator were invertible, cancel it to produce a smaller annihilating polynomial.

Proof. 1. If $Tv = \lambda v$ with $v \neq 0$, then $0 = m_T(T)v = m_T(\lambda)v$, so $m_T(\lambda) = 0$.
2. If $m_T(\lambda) = 0$, factor $m_T = (x - \lambda)q$. If $T - \lambda I$ were invertible, $(T - \lambda I)q(T) = 0$ would imply $q(T) = 0$, contradicting minimality.
3. Week 6 equates noninvertibility of $T - \lambda I$ with λ being an eigenvalue.
4. Over $\mathbb{C}$, factor m_T by the fundamental theorem of algebra from Week 6, whose analytic input was the cited extreme-value theorem. The preceding equivalence identifies every root as an eigenvalue, giving the product.

□

Proposition (Notes Proposition 67; LADR 5.22)

For finite-dimensional V, $\deg m_T \leq \dim V$.

Where the proof comes from

Want. Improve the operator-space bound $(\dim V)^2$ to $\dim V$.
Know. Vector iterates become dependent within $\dim V + 1$ terms, and invariant subspaces have quotient operators (Week 5).
First try. A relation on one vector need not hold on all vectors: $(T - I)e_1 = 0$ for $T = \operatorname{diag}(1, 2)$, but $T - I \neq 0$.
Key step. Use its nonzero invariant kernel and multiply annihilating polynomials for that kernel and its quotient.

Proof. 1. Induct on $n = \dim V$; for $n = 0$, $m_T = 1$ has degree 0. For $n > 0$, choose $v \neq 0$. A dependence of $v, Tv, \dots, T^n v$ gives nonzero q of degree at most n with $q(T)v = 0$.

2. Set $U = \ker q(T)$, a nonzero invariant subspace by Week 6. If $U = V$, Proposition 65 gives $m_T \mid q$, proving the bound.
3. Otherwise, $0 < \dim U < n$. Induction gives minimal polynomials r for $T|_U$ and s for T/U with $\deg r \leq \dim U$, $\deg s \leq \dim(V/U)$.
4. The equation $s(T/U) = 0$ means $s(T)V \subseteq U$; hence $r(T)s(T) = 0$. Proposition 65 gives $m_T \mid rs$.
5. Consequently $\deg m_T \leq \deg r + \deg s \leq \dim U + \dim(V/U) = n$, by Week 5.

□

Example (Notes Example 58) 1. On any nonzero space the identity has minimal polynomial $x - 1$.

2. For the length-n shift of Example 55, $T^n = 0$ but $T^{n-1}v_n = v_1 \neq 0$. In fact $v_n, Tv_n, \dots, T^{n-1}v_n$ is a basis, so no polynomial of degree below n annihilates T; therefore $m_T = x^n$.
3. For fixed dimension $n > 0$, every degree $1, \dots, n$ occurs.
4. Degree 1 is realized by I. For $2 \leq d \leq n$, take the direct sum of a length-$(d-1)$ shift and an identity on a space of dimension $n - d + 1$.
5. An annihilating polynomial must be divisible by x^{d-1} on the shift summand and by $x - 1$ on the identity summand, by Proposition 65. Writing that polynomial as $x^{d-1}q$, evaluation at 1 gives $q(1) = 0$, hence $x - 1 \mid q$ by the Week 4 factor identity. The product also annihilates the direct sum. Thus its minimal polynomial has degree d.

Find the first relation among matrices

1. For $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, try $A = aI$: the off-diagonal entry 1 makes this impossible.
2. Next solve $A^2 = aI + bA$ entrywise. Off-diagonal entries give $b = 4$, then diagonal entries give $a = -3$. The first relation yields $m_A = x^2 - 4x + 3$.

Definition (LADR 5B: cyclic vector)**.** A vector v is cyclic for T on an n-dimensional space if $v, Tv, \dots, T^{n-1}v$ is a basis.

Proposition (LADR 5B: cyclic-vector computation)
If $v, Tv, \dots, T^{n-1}v$ is a basis and $T^n v = -\sum_{j<n} a_j T^j v$, then $m_T = x^n + \sum_{j<n} a_j x^j$.

Where the proof comes from

Want. Replace n^2 matrix equations by n vector equations.
Know. The displayed iterates form a basis and every polynomial in T commutes with T.
Key step. A relation on v then holds on every basis iterate.

Proof. 1. For the displayed polynomial p, $p(T)v = 0$, hence $p(T)T^j v = T^j p(T)v = 0$ for every basis vector. Thus $p(T) = 0$.

2. No lower-degree nonzero polynomial can annihilate v, by independence of the basis iterates. Therefore no such polynomial annihilates T and $p = m_T$.

□

Example (LADR 5.26, 5.28)
For $C = \left(\begin{smallmatrix} 0 & 0 & 0 & 0 & -3 \\ 1 & 0 & 0 & 0 & 6 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{smallmatrix}\right)$, $e_1, Ce_1, \dots, C^4 e_1 = e_1, e_2, \dots, e_5$ and $C^5 e_1 = -3e_1 + 6Ce_1$. The cyclic computation gives $m_C = x^5 - 6x + 3$. Its eigenvalues are its roots by Proposition 66; exact radical expressions are not needed.

Proposition (LADR 5.31–5.32; Notes Example 60, divisibility step)

Restriction and quotient. If U is invariant, the minimal polynomials of $T|_U$ and T/U divide m_T. **Invertibility and inverse.** T is invertible if and only if $m_T(0) \neq 0$. In that case T^{-1} is a polynomial in T.

Where the proof comes from

Plan. First pass an annihilating relation to invariant parts; then solve that relation for the inverse.

Want. Transfer polynomial information to invariant parts and compute inverses.

Know. An annihilating polynomial restricts and passes to the quotient; zero roots are zero eigenvalues.

Key step. Apply Proposition 65 on each space and rearrange the constant term in $m_T(T) = 0$.

Proof. 1. Restriction and quotient evaluation of m_T are zero, because they are obtained from $m_T(T) = 0$. Proposition 65 on each space proves divisibility.

2. Proposition 66 and Week 6 show $m_T(0) = 0$ precisely when 0 is an eigenvalue, precisely when T is not invertible.
3. For nonzero V, write $m_T = x^d + a_{d-1}x^{d-1} + \cdots + a_1 x + a_0$ with $a_0 \neq 0$. Rearranging gives $T[-a_0^{-1}(T^{d-1} + a_{d-1}T^{d-2} + \cdots + a_1 I)] = I$. This commuting polynomial is T^{-1}. On $V = \{0\}$, $I = 0$ is its own inverse, also a polynomial.

□

Shortcuts

- A cyclic vector turns minimal-polynomial computation into one vector relation, because the relation propagates to its basis iterates.
- For inverses, isolate the nonzero constant term in $m_T(T) = 0$.
- A root must belong to the chosen field to be an eigenvalue; apply Proposition 66 over that field.

Traps

- An arbitrary annihilating polynomial can have extra roots: $(x-1)(x-2)$ annihilates I, but 2 is not its eigenvalue.
- A relation on one vector need not be a global relation: for $T = \operatorname{diag}(1, 2)$, $(T - I)e_1 = 0$ but $T - I \neq 0$.
- The zero-space convention matters: its minimal polynomial is 1, rather than $x - 1$.

§5 Distinct roots and diagonalization

The question. Can a polynomial relation certify an eigenbasis without finding eigenvectors? If $T^2 = I$ over $\mathbb{C}$, then the annihilating polynomial $(x-1)(x+1)$ has distinct roots.

Proposition (Notes Proposition 68; LADR 5.62)

An operator is diagonalizable over F if and only if its minimal polynomial splits into distinct linear factors over F. For the zero space, the empty product 1 satisfies this criterion.

Where the proof comes from

Want. Convert a polynomial relation into an eigenspace decomposition.

Know. Proposition 65 characterizes annihilating polynomials and Proposition 62 tests diagonalization.

Key step. Construct polynomials that equal 1 at one root and 0 at every other root.

Proof. 1. If T has an eigenbasis, let $\lambda_1, \dots, \lambda_r$ be its distinct eigenvalues. The product $p = \prod_j (x - \lambda_j)$ annihilates every eigenvector, hence T. Thus $m_T \mid p$ by Proposition 65, and Proposition 66 ensures all these roots occur in m_T; monicity gives $m_T = p$.

2. Conversely suppose $m_T = \prod_j (x - \lambda_j)$ with distinct roots. Define $p_j(x) = \prod_{k \neq j} (x - \lambda_k)/(\lambda_j - \lambda_k)$.

3. The polynomial $\sum_j p_j - 1$ has degree at most $r - 1$ and vanishes at the r roots, so it is zero by the Week 4 root bound. Hence $\sum_j p_j(T) = I$.

4. Since $(x - \lambda_j)p_j$ is a scalar multiple of m_T, its evaluation is zero. Thus $p_j(T)v \in E(\lambda_j, T)$ for each v.

5. Every $v = \sum_j p_j(T)v$ therefore belongs to the sum of eigenspaces. Proposition 62 proves diagonalizability. The zero-space case follows from its empty eigenbasis.

□

Definition (Notes Chapter 13: repeated roots and greatest common divisor)**.** A root λ has multiplicity k if $p = (x - \lambda)^k q$ and $q(\lambda) \neq 0$. A polynomial is squarefree if no root is repeated over $\mathbb{C}$. For $p = \sum_j a_j x^j$, its formal derivative is $p' = \sum_{j \geq 1} j a_j x^{j-1}$. The monic greatest common divisor $\gcd(p, q)$ divides both polynomials and is divisible by every common divisor.

Proposition (Notes Chapter 13: derivative test)

For a nonconstant complex polynomial p, its roots are distinct if and only if $\gcd(p, p') = 1$. Thus over $\mathbb{C}$, T is diagonalizable exactly when $\gcd(m_T, m_T') = 1$ (with the constant case automatic).

Where the proof comes from

Want. Test repeated roots without solving the polynomial.

Know. Week 6 factors complex polynomials into linear factors; division gives the Euclidean algorithm.

Key step. Differentiate a root factorization and track whether the root remains.

Proof. 1. If $p = (x - \lambda)^k q$ with $q(\lambda) \neq 0$, then $p' = k(x - \lambda)^{k-1} q + (x - \lambda)^k q'$. For $k = 1$, $p'(\lambda) = q(\lambda) \neq 0$; for $k > 1$, $p'(\lambda) = 0$.

2. Hence common roots of p, p' are exactly repeated roots. A nonconstant common factor has a root by Week 6, so there are no common roots exactly when the monic gcd is 1.

3. Compute the gcd by repeated polynomial division: replacing (p, q) by $(q, p - hq)$ preserves common divisors; degree decreases until a zero remainder, whose predecessor normalized to be monic is the gcd. Apply Proposition 68.

□

Where the proof comes from

Want. Decide diagonalizability from the periodicity relation $T^n = I$.
Know. The derivative test detects repeated roots; minimal polynomials divide annihilating polynomials.
Key step. Compare $x^n - 1$ with its simple derivative.

Example (Notes Example 59)

Over $\mathbb{C}$, suppose $T^n = I$ for a positive integer n.

1. Proposition 65 gives $m_T \mid x^n - 1$. The derivative of $x^n - 1$ is nx^{n-1}; its only possible root is 0, which is not a root of $x^n - 1$.
2. The derivative test therefore gives distinct roots for $x^n - 1$, hence for its divisor m_T. Proposition 68 gives an eigenbasis.

Where the proof comes from

Want. Pass diagonalizability to an invariant restriction or quotient.
Know. Their minimal polynomials divide the original one.
Key step. Divisibility preserves splitting into distinct linear factors.

Example (Notes Example 60; LADR 5.65)

If T is diagonalizable and U invariant, then $T|_U$ and T/U are diagonalizable.

1. Their minimal polynomials divide m_T, by the restriction and quotient result already proved. Proposition 68 gives distinct linear factors for m_T.
2. Each divisor is again a product of distinct linear factors (remove a subset of those factors). Apply Proposition 68 on each space.

Apply the derivative test to a companion matrix (LADR 5.60)

1. For the previous companion matrix, $p = x^5 - 6x + 3$ and $p' = 5x^4 - 6$. If $p(z) = p'(z) = 0$, then $z^4 = 6/5$ and $p(z) = -(24/5)z + 3 = 0$, so $z = 5/8$.
2. But $(5/8)^4 = 625/4096 \neq 6/5$ since $3125 \neq 24576$. Thus no common root exists, and the matrix is diagonalizable over $\mathbb{C}$. This gives an exact justification without relying on approximate roots.

Detect a repeated factor (LADR 5.61)

1. Over $\mathbb{R}$ or $\mathbb{C}$, for $B = \left(\begin{smallmatrix} 6 & 3 & 4 \\ 0 & 6 & 2 \\ 0 & 0 & 7 \end{smallmatrix}\right)$, the eigenvalues are 6 and 7. Proposition 67 bounds $\deg m_B \leq 3$, and Proposition 66 requires both roots.
2. Compute $(B - 6I)(B - 7I) = \left(\begin{smallmatrix} 0 & -3 & 6 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{smallmatrix}\right) \neq 0$, whereas $(B - 6I)^2(B - 7I) = 0$. Hence Proposition 65 gives $m_B = (x - 6)^2(x - 7)$, and Proposition 68 rules out diagonalization.

Shortcuts

- Use the Euclidean algorithm on m_T, m_T' over $\mathbb{C}$ to avoid computing roots.
- If an annihilating polynomial splits into distinct factors, so does m_T by divisibility, immediately proving diagonalizability.

Traps

- Squarefree alone is insufficient over $\mathbb{R}$: $x^2 + 1$ is squarefree but does not split, and the quarter-turn is not diagonalizable over $\mathbb{R}$.
- An annihilating polynomial with repeated roots does not rule out an eigenbasis: $(x - 1)^2$ annihilates I. The criterion concerns the minimal polynomial.

§6 Further eigenvalue tests from the textbook

The question. Can one guarantee or locate eigenvalues without solving their polynomial? Every real operator in odd dimension has a real eigenvalue, and a matrix's row sizes bound where its eigenvalues can lie.

Proposition (LADR 5.33: even quadratic kernel)

For an operator T on a finite-dimensional real space and real b, c with $b^2 < 4c$, the dimension of $\ker(T^2 + bT + cI)$ is even.

Where the proof comes from

Want. Prove a parity restriction using invariant subspaces.
Know. $x^2 + bx + c = (x + b/2)^2 + c - b^2/4$ has no real root.
Key step. On its invariant kernel, every new nonzero vector generates a two-dimensional invariant span.

Proof. 1. Restrict T to $K = \ker(T^2 + bT + cI)$, invariant by Week 6. On K, $T^2 + bT + cI = 0$, so no nonzero vector is a real eigenvector, by substitution and positivity of the quadratic.
2. Choose an invariant subspace $U \subseteq K$ of largest even dimension; 0 is available. If $w \in K \setminus U$, the space $W = \operatorname{span}(w, Tw)$ is invariant by the quadratic relation, and has dimension 2 because w is not an eigenvector.
3. The invariant intersection $U \cap W$ cannot have dimension 1 (that would be an invariant line), or 2 (then $w \in U$). Thus it is zero.
4. Then $U + W$ is invariant of even dimension $\dim U + 2$, contradicting maximality. Therefore $U = K$ and $\dim K$ is even.

□

Proposition (Real polynomial factorization; LADR 4.16)

Every nonconstant real polynomial is a product of a nonzero real constant, real linear factors, and real quadratic factors with negative discriminant.

Where the proof comes from

Want. Obtain the real factors needed in the odd-dimension argument.
Know. Week 6 factors polynomials over $\mathbb{C}$.
Key step. Pair every nonreal root with its conjugate.

Proof. 1. For real coefficients, conjugating the finite sum gives $\overline{p(z)} = p(\overline{z})$. Thus nonreal roots occur in conjugate pairs, with the same multiplicity, by conjugating their factorization.
2. For $z = a + bi$ with $b \neq 0$, the pair contributes $(x - z)(x - \overline{z}) = x^2 - 2ax + a^2 + b^2$, whose discriminant is $-4b^2 < 0$.
3. Group these pairs and retain the real roots as linear factors in the complex factorization from Week 6. The leading real coefficient is the remaining constant.

□

Proposition (LADR 5.34: odd-dimensional real operators)

Every operator on a real vector space of positive odd finite dimension has an eigenvalue.

Where the proof comes from

Want. Force a real eigenvalue even though a real polynomial need not have a real root.
Know. Real factorization gives linear or irreducible quadratic factors; the preceding result gives even quadratic-kernel dimension.
Key step. A quadratic factor produces a proper invariant range of smaller odd dimension; induct there.

Proof. 1. Induct on odd n. For $n = 1$, every operator is scalar. If m_T has a real linear factor, Proposition 66 gives an eigenvalue.
2. Otherwise real factorization supplies $m_T = qh$ with $q = x^2 + bx + c$, $b^2 < 4c$. Put $W = \operatorname{range} q(T)$, invariant by Week 6.
3. If $W = V$, $h(T)q(T) = 0$ would imply $h(T) = 0$, contradicting minimality. Thus W is proper.
4. Rank-nullity and the even-kernel result give $\dim W = n - \dim \ker q(T)$ odd, hence positive. It is less than n, so induction gives a real eigenvector of $T|_W$, also one of T.

□

Definition (LADR 5.66: Gershgorin disks)**.** For $F = \mathbb{R}$ or $\mathbb{C}$ and a matrix $A \in F^{n\times n}$, its ith Gershgorin disk is $D_i = \{z \in F : |z - a_{ii}| \le \sum_{j\neq i} |a_{ij}|\}$. Over $\mathbb{R}$ these sets are intervals; to locate all complex eigenvalues of a real matrix, regard that matrix over $\mathbb{C}$.

Proposition (LADR 5.67: Gershgorin disk theorem)
Every eigenvalue of A lies in $\bigcup_i D_i$.

Where the proof comes from

Want. Locate eigenvalues from row entries.
Know. An eigenvector has a coordinate of maximum nonzero modulus.
Key step. Use that row of $Ax = \lambda x$ and divide by its coordinate.

Proof. 1. Choose an eigenvector $x \neq 0$ and an index i with $|x_i| = \max_j |x_j| > 0$. The ith row gives $(\lambda - a_{ii})x_i = \sum_{j\neq i} a_{ij}x_j$.
2. The triangle inequality yields $|\lambda - a_{ii}| \le \sum_{j\neq i} |a_{ij}|\,|x_j|/|x_i| \le \sum_{j\neq i} |a_{ij}|$. Thus $\lambda \in D_i$.

□

Bound eigenvalues by row sizes

1. For $A = \left(\begin{smallmatrix} 2 & 1 & 0 \\ 0 & 5 & 3 \\ 0 & 0 & 8 \end{smallmatrix}\right)$, the three disks have centers $2, 5, 8$ and radii $1, 3, 0$. Thus each eigenvalue belongs to $\{|z - 2| \le 1\} \cup \{|z - 5| \le 3\} \cup \{8\}$.
2. The triangular test gives the actual eigenvalues $2, 5, 8$, and each lies in the stated union. Gershgorin provides bounds for matrices that are not triangular as well.

Shortcuts

- A real odd-dimensional operator has a real eigenvalue without any root calculation, by the parity proof.
- Gershgorin bounds require only diagonal entries and sums of off-diagonal absolute values.

Traps

- Even real dimension need not give a real eigenvalue: the quarter-turn on $\mathbb{R}^2$ has none.
- Disk membership alone does not characterize eigenvalues: $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ has unit disks containing 0, but its only eigenvalues are ± 1.

§7 Problem set

The question. Find minimal polynomials by the first operator relation, and use their roots and multiplicities to decide diagonalizability. For example, $A^2 = -I$ with nonscalar A on F^2 gives $m_A = x^2 + 1$.

HW7 P1. Let $A = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. (a) Find m_A over $\mathbb{R}$ and $\mathbb{C}$. (b) Find m_B over $\mathbb{R}$. (c) Recover the diagonalizability conclusions of HW6 using Proposition 68.
Hint: Use the first-relation method: square A and $B - I$, then rule out scalar matrices.

HW7 P2. For differentiation $D : \mathcal{P}_3(\mathbb{R}) \to \mathcal{P}_3(\mathbb{R})$, find its minimal polynomial and decide diagonalizability.
Hint: Use the first-relation method on the iterates of x^3.

HW7 P3. Let T be invertible on a finite-dimensional space and $m_T = x^d + a_{d-1}x^{d-1} + \cdots + a_1 x + a_0$. (a) Prove $a_0 \neq 0$. (b) Prove T^{-1} is a polynomial in T. (c) Over $\mathbb{C}$, if $m_T = x^4 + x^3 - x^2 - 5x + 4$, express T^{-1} explicitly, find the eigenvalues, and decide diagonalizability.
Hint: Use LADR 5.32 to isolate $a_0 I$; apply Propositions 66 and 68 after factoring.

HW7 P4 (optional). Prove $m_{T-\lambda I}(x) = m_T(x + \lambda)$. If $c \neq 0$ and $d = \deg m_T$, prove $m_{cT}(x) = c^d m_T(x/c)$.
Hint: Use the minimality definition after affine substitution, then reverse the substitution to exclude smaller degree.

HW7 P5. Let A be a two-by-two complex matrix with only eigenvalue λ. Prove m_A is either $x - \lambda$ or $(x - \lambda)^2$; prove the first case occurs exactly when $A = \lambda I$, and conclude that A is diagonalizable exactly in that case.
Hint: Combine complex factorization, Propositions 66–68, and the degree bound.

HW7 P6. For $C = \left(\begin{smallmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{smallmatrix}\right)$, find eigenvalues, eigenspaces and minimal polynomial, and decide diagonalizability.
Hint: Use Proposition 60 and kernel equations, then Proposition 65 to test $(C - 2I)(C - 3I)$.

HW7 P7 (optional). Suppose $m_T = \prod_{j=1}^{r}(x - \lambda_j)$ with distinct roots. Construct p_j with $p_j(\lambda_j) = 1$, $p_j(\lambda_k) = 0$ for $k \neq j$, and show that $p_j(T)$ projects onto $E(\lambda_j, T)$ along the other eigenspaces.
Hint: Use the interpolation polynomials in the proof of Proposition 68.

Exam practice (midterm2-review.pdf, Problem 6; Axler 5C.7). For T on finite-dimensional V and $v \in V$, prove that there is a unique monic polynomial p_v of smallest degree with $p_v(T)v = 0$, and prove $p_v \mid m_T$.
Hint: Use the first-dependence argument and polynomial division on m_T by p_v; when $v = 0$, use $p_v = 1$.